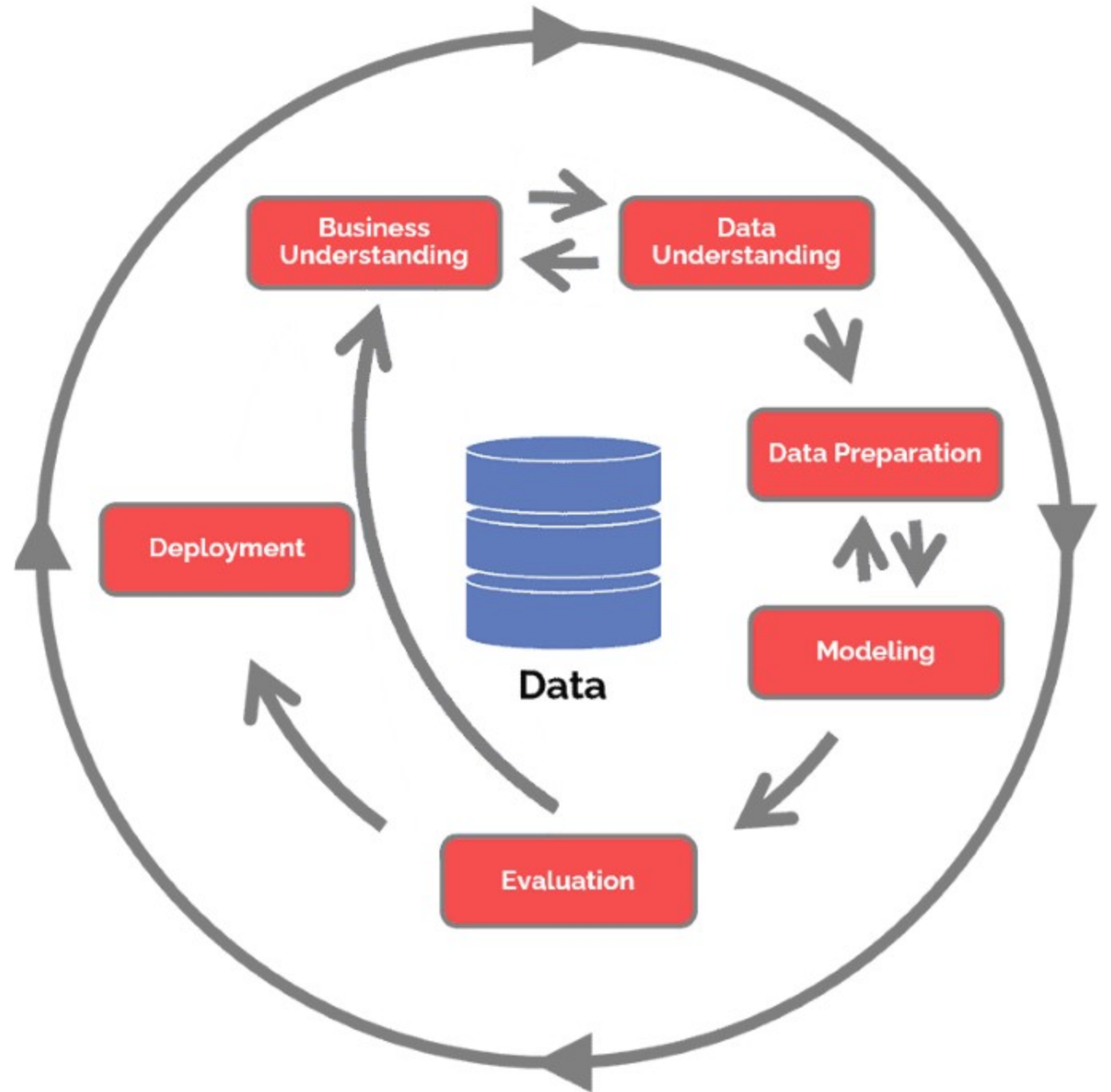




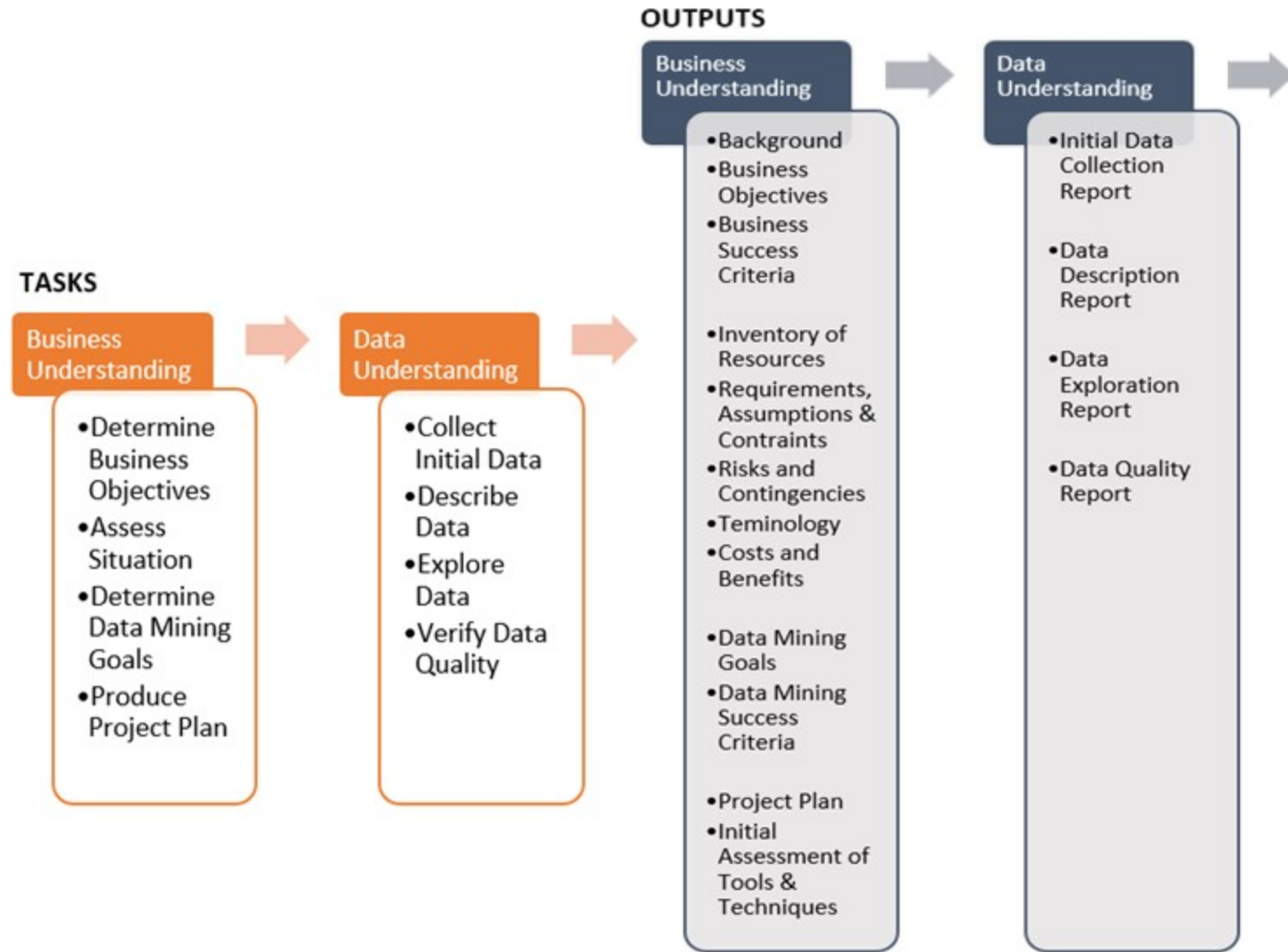
Producing Analytic Plans

CRISP-DM

Cross Industry
Standard Process for
Data Mining



CRISP-DM: Tasks and Outputs



Before Data Prep...

1

Identify research questions and/or hypotheses.

2

Select and access a dataset.

3

List inclusion/exclusion criteria.

4

Review the data to determine the variables to be used in the main analysis.

5

Select the appropriate statistical methods and software.



AND...

- Create table shells.
- Create a data dictionary.
- Create a working copy of the dataset.
- Clean the data in the working file.
- Create an analysis file.

Analysis Plan

The analysis plan usually contains:

1. Research question(s) and/or hypotheses

Analysis Plan

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1. Research question(s) and/or hypotheses
2. Dataset(s) to be used

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 - Example: the main exposure, outcome, and stratifying variables
5. Statistical methods and software to be used
6. Key table shells
 - Example: univariable, bivariable, and stratified

Table Shells

Education Level of Adults

Univariate Table

Highest Educational Achievement	Number	Percent
None		
Primary school		
Secondary school		
Post-secondary school		
Total		

Bivariate Table

Exposure Variable: <u>Highest Educational Achievement</u>	Outcome Variable: <u>Tobacco use</u>				
	Yes	No	Total	PR (95% CI)	POR (95% CI)
None				1.0	1.0
	(%)	(%)			
Primary school					
	(%)	(%)			
Secondary school					
	(%)	(%)			
Post-secondary school (referent)					
	(%)	(%)			
Total					
$\chi^2 = \text{____}, df = \text{____}, p = \text{____}$					

Multivariate Table

Exposure Variable:	Outcome Variable: <u>Tobacco use</u>				
<u>Highest Educational achievement</u>	Yes	No	Total	PR (95% CI)	POR (95% CI)
None				1.0	1.0
	(%)	(%)			
Male					
	(%)	(%)			
Female (referent) ¹⁵					
	(%)	(%)			
Primary school				1.0	1.0
	(%)	(%)			
Male					
	(%)	(%)			
Female (referent)					
	(%)	(%)			
Secondary school				1.0	1.0
	(%)	(%)			
Male					
	(%)	(%)			
Female (referent)					
	(%)	(%)			
Post-secondary school				1.0	1.0
	(%)	(%)			
Male					
	(%)	(%)			
Female (referent)					
	(%)	(%)			
Total					
	(%)	(%)			
$\chi^2 = \text{____}, df = \text{____}, p = \text{____}$					

Questions for your Dataset/s

- Who owns the database?
- How can you get access to the database?
- Do you need permission to use the database?
- Does the database cost anything to use?
- Are there rules about storing the database?
- What was the purpose of the study?
- What are the study hypotheses?
- What methods were used to identify (select) the population under study and gather information from them?
- How was the data collected, entered, and checked for quality control?
- In what program (e.g., MS Access/Epi Info, Excel, SQL, etc.) and in what format is the dataset stored (e.g., text, ASCII, comma-delimited, etc.)?
- How many records are in the dataset?
- Were weights used?